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Topic

- My research topic focuses on the effects of computer games on players, particularly in three main aspects: addiction, aggression and cognitive influence. Video games have become a major part of today's entertainments. with millions of people playing daily. However, there has been controversy over whether the video games will have negative psychological and behavioural effects on players, especially for children and adolescents(Király et al., 2018; Przybylski & Weinstein, 2019).
- One of the issues is that the gaming addiction. Some people spend a excessive amount of time on video games which caused negative impact on their daily life, academic performances and mental health. Some games researchers believed that gaming addiction is similar with gambling addiction, since both involve compulsive behavior and reward-seeking mechanisms(King & Delfabbro, 2018; Pontes et al., 2021).
- Another concern is the link between violent video games and aggressive behavior. Some research revealed that touching violent video games might make players increasing aggressive thoughts or desensitized to violence (Prescott et al., 2018). However, there are some researches give opposite opinion, which there is no direct evidence proving there is a relationship between games and real-world aggression (Johannes et al., 2022).
- Finally, I want to discuss about how video games impact our cognitive ability, such as reaction time, the ability of solving problems and making decision. There are some researches believed that frequently gaming can strengthen people's cognitive flexibility and multitasking skills (Bediou et al., 2018), while others worry that excessive gaming might impair focus and social interactions (Li et al., 2021).
- The purpose of this research is to use academic researches and real-world examples to examine the positive and negative influences caused by video games, helping players to understand whether gaming is more harmful or beneficial.

Initial Query

- As previous talk in #Topic part, I prepare to explore the effects of video games focusing on three key parts: video game addiction, the impact of violent games on aggression and the cognitive effects of gaming. Since currently video games are widely played, I want to discuss and find both their potential benefits and harms by searching researches that examined compulsive behaviour, psychological aggression and cognitive impacts among players.

- First, I will search research papers from 2015 onward using academic databases such as Google Scholar, UTSC library, PsycINFO, ACM Digital Library, IEEE Xplore, and PubMed. I use specific terms which are relevant to my topic to capture a wide range of studies. Here are some examples of my search terms:
 - "video game addiction" AND "compulsive gaming" AND "mental health"
 - "gaming disorder" AND "WHO classification" AND "behavioural effects"
 - "violent video games" AND "aggression" AND "meta-analysis"
 - "first-person shooter games" AND "real-world violence" AND "desensitization"
 - "video games" AND "cognitive flexibility" AND "reaction time improvement"
 - "gaming" AND "attention span" AND "cognitive benefits OR cognitive impairment"
- After searching the relevant papers, I will filter out outdated studies, non-peer-reviewed articles, and research with small sample sizes. This action can help me to get high-quality sources and precise conclusion with my topic.

A List of Your Chosen Databases

1. Google Scholar
2. UTSC Library
3. PsycINFO (via University of Toronto Library)
4. ACM Digital Library
5. IEEE Xplore
6. PubMed
7. PMC

Actually, I mainly use the Google Scholar and UTSC Library. Others are not used yet but I record them into this list to make them as an additional resource. Maybe later I will find some relevant helpful sources on them.

Query (Google Scholar)

- **Date of Query:** February 1, 2025
- **Search Query:**
 - "video game addiction" AND "compulsive gaming" AND "mental health" after:2015
 - "violent video games" AND "aggression" AND "meta-analysis" after:2015
 - ("Gaming Disorder" OR "Problematic Gaming") AND "Mental Health Effects"
 -
- **Filters Applied:**
 - **Year:** 2015–Present
 - Only peer-reviewed journal articles

- Sort by relevance and citation count

List of Papers & Full Bibliography for the Retrieved Papers

1. The Effects of Video Games on Cognition and Brain Structure: Potential Implications for Neuropsychiatric Disorders

- **Abstract** : There has been interest in the effect of video games on brain structure and cognitive function in both experimental and therapeutic contexts. Overall, there is evidence that specific types of video games can alter brain structure or improve certain aspects of cognitive functioning. Video games can also be useful as neuropsychological assessment tools. While research in this area is still at a very early stage, there are interesting results that encourage further work in this field, and hold promise for utilizing this technology as a powerful therapeutic and experimental tool.
- **APA citation** : Shams, T. A., Foussias, G., Zawadzki, J. A., Marshe, V. S., Siddiqui, I., Müller, D. J., & Wong, A. H. C. (2015). The effects of video games on cognition and brain structure: Potential implications for neuropsychiatric disorders. *Current Psychiatry Reports*, 17(9), 71.

2. Effects of Violent Video Games on Aggressive Cognition and Aggressive Behavior

- **Abstract** : In this study, we examined how playing a violent video game affected aggressive cognition and aggressive behavior. A total of 300 children ($M_{age} = 6.38$, $SD = 0.25$) were randomly recruited to play a violent or a nonviolent video game. Results revealed that briefly exposing children to a violent video game increased aggressive cognition and aggressive behavior. In addition, a significant game $\times$ sex interaction showed that this effect was larger for boys than for girls. Mediation pathways were found such that aggressive cognition mediated the relationship between violent video games and aggressive behavior, especially for boys. Findings were interpreted within and supported the framework of the general aggression model. Violent video game effects remain a societal concern, and boys should be regarded as a special group for aggression intervention.
- **APA citation** : Zhang, Q., Cao, Y., & Tian, J. (2021). Effects of violent video games on aggressive cognition and aggressive behavior. *Cyberpsychology, Behavior, and Social Networking*, 24(1), 5-10.

3. Mindfulness-Oriented Recovery Enhancement for Video Game Addiction in Emerging Adults: Preliminary Findings from Case Reports

- **Abstract** : Video game addiction is increasingly prevalent in emerging adults and is associated with physical and psychosocial impairments. However, few evidence-based treatments for video game addiction have been identified or evaluated.

Mindfulness treatment is effective for substance use disorders and behavioral addictions, and may be a promising intervention for video game addiction. This report presents theoretical support for mindfulness treatment of video game addiction, describes an adapted mindfulness treatment (i.e., Mindfulness-Oriented Recovery Enhancement (MORE)) for video game addiction, and presents case vignettes that describe preliminary effects of MORE in reducing video game addiction and increasing positive coping. Important methodological issues including conceptualization of the treatment model, feasibility of treatment manual adaptation, and promising therapeutic benefits of the MORE intervention are discussed. Our experience developing a mindfulness treatment of video game addiction may be informative to other investigators planning similar studies. In addition, this report could provide useful guidance to clinicians.

- **APA citation :** Li, W., Garland, E. L., O'Brien, J. E., Tronnier, C., McGovern, P., Anthony, B., & Howard, M. O. (2018). Mindfulness-oriented recovery enhancement for video game addiction in emerging adults: Preliminary findings from case reports. *International Journal of Mental Health and Addiction*, 16, 928-945.

4. **Fight fire with rainbows: The effects of displayed violence, difficulty, and performance in digital games on affect, aggression, and physiological arousal**

- **Abstract :** There is a large amount of variables that need to be taken into account when studying the effects of violent content in digital games; one of those being difficulty. In the current study participants played a modified first-person shooter in one of four different conditions, with either high or low difficulty and high or low violent game content. We assessed number of kills and number of deaths as game performance. Neither the difficulty nor the displayed violence had an effect on psychophysiological arousal during play, post-game aggressive cognitions, nor **aggressive behavior**. Thus, this study corroborates previous research indicating that violence in games does not substantially influence human behavior or experience, and other game characteristics deserve more attention in game effects studies. In addition, findings showed that challenge manipulated by game difficulty is of main importance for post-game emotions: Number of deaths predicted positive affect, but only in the low difficulty condition while number of kills was a positive predictor for positive affect and a negative predictor for negative affect.
- **APA citation :** Kneer, J., Elson, M., & Knapp, F. (2016). Fight fire with rainbows: The effects of displayed violence, difficulty, and performance in digital games on affect, aggression, and physiological arousal. *Computers in human behavior*, 54, 142-148.

5. **Violent Video Games: The Effects of Narrative Context and Reward Structure on In-Game and Postgame Aggression**

- **Abstract :** This research examines how narrative context and reward structures in violent video games affect aggression during and after gameplay. Results indicate

that games with antisocial narratives and reward systems promoting violence increase aggressive thoughts and behaviors.

- **APA citation** : Sauer, J. D., Drummond, A., & Nova, N. (2015). Violent video games: The effects of narrative context and reward structure on in-game and postgame aggression. *Journal of Experimental Psychology: Applied*, 21(3), 205–214.

6. The Internet and Aggression: Motivation, Disinhibitory, and Opportunity Aspects

- **Abstract** : This theoretical paper explores how the internet facilitates aggressive behaviors through motivational factors, disinhibition, and increased opportunities for aggression. It discusses the role of online anonymity and the prevalence of aggressive content.
- **APA citation** : Malamuth, N., Linz, D., & Weber, R. (Year). The internet and aggression: Motivation, disinhibitory, and opportunity aspects. In *The Social Net: Understanding Our Online Behavior* (pp. X–Y).

7. Violent Video Game Effects on Salivary Cortisol, Arousal, and Aggressive Thoughts in Children

- **Abstract** : An experiment investigated the effects of violent content in video games on two physiological indicators of the fight-or-flight response (cortisol and cardiovascular changes) and on accessibility of aggressive thoughts in children. Participants played a randomly assigned violent or nonviolent video game, rated the game on several dimensions, and did a word completion task. Results showed that the **violent video game** increased cortisol and (for boys) cardiovascular arousal (relative to baseline) more than did the equally exciting nonviolent game. The violent game also increased the accessibility of aggressive thoughts. The cortisol findings in particular suggest that playing a **violent video game** may activate the **sympathetic nervous system** and elicit a fight-or-flight type response in children. Theoretical implications and future research are discussed.
- **APA citation** : Gentile, D. A., Bender, P. K., & Anderson, C. A. (2017). Violent video game effects on salivary cortisol, arousal, and aggressive thoughts in children. *Computers in Human Behavior*, 70, 39-43.

8. Video Game Addiction: What Can We Learn from a Media Neuroscience Perspective?

- **Abstract** : In recent years, video game addiction has received considerable empirical attention. Unfortunately, this research is stymied by inconsistencies in both conceptual and operational definitions of video game addiction. Moreover, the use of several video game addiction scales makes it difficult to estimate the prevalence and potential effects of video game addiction. While game genre is often treated as a predictor of video game addiction, existing measures often downplay the structural and social characteristics of video games that may contribute to behavioral outcomes such as increased playing time and addiction. In an effort to provide the clarity

necessary to overcome these issues, we review research on video game addiction with a focus on the largely ignored unique characteristics of video games that are crucial for a more complete conceptualization of video game addiction. With this review in mind, we offer a conceptual framework for the integration of video game addiction within the broader context of behavioral addictions. Finally, we consider the neurological foundation of addiction and suggest opportunities for media neuroscientists to increase understanding and prediction of video game addiction and explore how game content features interact with reward systems in the brain.

- **APA citation** : Craighead, B., Huskey, R., & Weber, R. (2015). Video Game Addiction: What can we learn from a media neuroscience perspective?. *Revista Argentina de Ciencias del Comportamiento*, 7(3), 00-00.

9. **The Association between Video Game Time and Adolescent Mental Health: Evidence from Rural China**

- **Abstract** : As digital devices like computers become more widely available in developing countries, there is a growing need to understand how the time that adolescents spend using these devices for recreational purposes such as playing video games is linked with their mental health outcomes. We measured the amount of time that adolescents in rural China spent playing video games and the association of video game time with their mental health. We collected data from primary and junior high schools in a poor, rural province in northwest China (n = 1603 students) and used the Depression, Anxiety, and Stress Scales (DASS-21) to measure mental health symptoms. The results indicated that the average video game time was about 0.69 h per week. There was a significant association between adolescent video game time and poorer mental health. Each additional hour of playing video games also increased the chance of having moderate or above symptoms. Moreover, boys and non-left-behind children had worse mental health if they played more video games. Our study contributes to literature on the links between recreational screen time and mental health, and it sheds light on an issue addressed by recent government legislation to limit the video game time of minors in China.
- **APA citation** : Li, L., Abbey, C., Wang, H., Zhu, A., Shao, T., Dai, D., ... & Rozelle, S. (2022). The association between video game time and adolescent mental health: evidence from Rural China. *International journal of environmental research and public health*, 19(22), 14815.

10. **Are Mental Health Effects of Internet Use Attributable to the Web-Based Content or Perceived Consequences of Usage? A Longitudinal Study of European Adolescents**

- **Abstract** : Adolescents and young adults are among the most frequent Internet users, and accumulating evidence suggests that their Internet behaviors might affect their mental health. Internet use may impact mental health because certain Web-based content could be distressing. It is also possible that excessive use, regardless

of content, produces negative consequences, such as neglect of protective offline activities.

- **APA citation** : Hökby, S., Hadlaczký, G., Westerlund, J., Wasserman, D., Balazs, J., Germanavicius, A., ... & Carli, V. (2016). Are mental health effects of internet use attributable to the web-based content or perceived consequences of usage? A longitudinal study of European adolescents. *JMIR Mental Health*, 3(3), e5925.

11. **The Relationship between Game Addiction and Aggression among Adolescents with Mediating Role of Narcissism and Self-Control**

- **Abstract** : With the increase of game addiction, the problem of aggression as one of the most serious issues confronting today's society is affecting the mental health of the young generation. While available research has clarified the relationship between game addiction and aggression, this research contributes to future literature through examining the mediating role of narcissism and self-control in this relationship.
- **APA citation** : Pan, Y., Motevalli, S., & Yu, L. (2024). The Relationship between Game Addiction and Aggression among Adolescents with Mediating Role of Narcissism and Self-Control. *Iranian Journal of Psychiatry*, 1-11.

12. **Does playing violent video games cause aggression? A longitudinal intervention study**

- **Abstract** : It is a widespread concern that violent video games promote aggression, reduce pro-social behaviour, increase impulsivity and interfere with cognition as well as mood in its players. Previous experimental studies have focussed on short-term effects of violent video gameplay on aggression, yet there are reasons to believe that these effects are mostly the result of priming. In contrast, the present study is the first to investigate the effects of long-term violent video gameplay using a large battery of tests spanning questionnaires, behavioural measures of aggression, sexist attitudes, empathy and interpersonal competencies, impulsivity-related constructs (such as sensation seeking, boredom proneness, risk taking, delay discounting), mental health (depressivity, anxiety) as well as executive control functions, before and after 2 months of gameplay. Our participants played the violent video game Grand Theft Auto V, the non-violent video game The Sims 3 or no game at all for 2 months on a daily basis. No significant changes were observed, neither when comparing the group playing a violent video game to a group playing a non-violent game, nor to a passive control group. Also, no effects were observed between baseline and posttest directly after the intervention, nor between baseline and a follow-up assessment 2 months after the intervention period had ended. The present results thus provide strong evidence against the frequently debated negative effects of playing violent video games in adults and will therefore help to communicate a more realistic scientific perspective on the effects of violent video gaming.
- **APA citation** : Kühn, S., Kugler, D., Schmalen, K. *et al.* Does playing violent video games cause aggression? A longitudinal intervention study. *Mol Psychiatry* 24, 1220–

Exclude Papers

1. **The Impact of Video Games on Motor Skills Development in Older Adults”**
 - Excluded because it focuses on elderly individuals rather than children or adolescents, which is the target population of my research.
2. **The Role of Virtual Reality in Physical Rehabilitation**
 - Excluded because it discusses the use of video games for rehabilitation rather than their psychological or behavioural effects.
3. **Video Games and Economic Productivity: A Macroeconomic Analysis**
 - Excluded because it analyzes economic impacts rather than cognitive, psychological, or behavioral effects.
4. **Gender Differences in Video Game Preferences and Playtime**
 - Excluded because it focuses on gender differences in gaming habits rather than the effects of gaming itself.

Reference for Topic part

1. Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills. *Psychological Bulletin*, 144(1), 77–110. <https://doi.org/10.1037/bul0000130>
2. Johannes, N., Vuorre, M., & Przybylski, A. K. (2022). Video game violence, aggression, and delinquency: A registered report. *Royal Society Open Science*, 9(1), 211108. <https://doi.org/10.1098/rsos.211108>
3. King, D. L., & Delfabbro, P. H. (2018). Video game addiction. *Annual Review of Clinical Psychology*, 14(1), 151–176. <https://doi.org/10.1146/annurev-clinpsy-032816-045209>
4. Király, O., Slecza, P., Pontes, H. M., Griffiths, M. D., & Demetrovics, Z. (2018). Validation of the Ten-Item Internet Gaming Disorder Test (IGDT-10) and evaluation of gaming patterns in a national sample of gamers. *Computers in Human Behavior*, 90, 38–46. <https://doi.org/10.1016/j.chb.2018.08.012>
5. Li, X., Beullens, K., & Eggermont, S. (2021). The effects of gaming disorder on adolescent sleep: A longitudinal study. *Sleep Health*, 7(2), 163–171. <https://doi.org/10.1016/j.jth.2020.11.005>
6. Pontes, H. M., Schivinski, B., Sindermann, C., Li, M., Becker, B., Zhou, M., & Montag, C. (2021). Measurement and conceptualization of Gaming Disorder according to the World Health Organization framework: The development of the Gaming Disorder Test. *International Journal of Mental Health and Addiction*, 19(2), 508–528. <https://doi.org/10.1007/s11469-019-00088-z>

7. Prescott, A. T., Sargent, J. D., & Hull, J. G. (2018). Metaanalysis of the relationship between violent video game play and physical aggression over time. *Proceedings of the National Academy of Sciences*, 115(40), 9882–9888.
<https://doi.org/10.1073/pnas.1611617114>
8. Przybylski, A. K., & Weinstein, N. (2019). Violent video game engagement is not associated with adolescents' aggressive behaviour: Evidence from a registered report. *Royal Society Open Science*, 6(2), 171474. <https://doi.org/10.1098/rsos.171474>